

Falcon: A High Performance, Lightweight Web Framework



Falcon is a lightweight, high performance framework for building cloud APIs. If your aim is to serve requests as quickly as possible by putting in some extra effort during the development stage, then you should give Falcon a try.

To build Web applications, developers generally tend to pick a language like PHP. While Python is closely associated with scientific computing, it offers an array of choices when it comes to building Web scale applications. The world of Python is loaded with Web frameworks (some of which are listed in Table 1).

Web framework	Description
Bottle	This is a fast and simple WSGI micro Web framework
Django	A Web framework for perfectionists with deadlines
Flask	This is a micro-framework based on Werkzeug and Jinja2
Pyramid	A small, fast, down-to-earth Web framework. It has been built as a part of the Pylons project
Torando	A Python Web framework and asynchronous networking library
Falcon	A high-performance Python framework for building cloud APIs
Wheezy Web	A high performance and high concurrency WSGI Web framework

Table 1: Web frameworks in Python

As can be seen from Table 1, each Web framework has its own set of features and target scenarios. If you are interested in a comparative analysis of their performance, visit <http://klen.github.io/py-frameworks-bench/>. Post analysis, Falcon will emerge as one among the top ranked frameworks.

Falcon design goals

Falcon official documentation lists the following three major design goals for Falcon (Figure 1).

Fast: One of the most important requirements of cloud APIs is that they should respond to requests as quickly as possible. This becomes a critical feature in real-time scenarios when the number of concurrent requests is high. Falcon is one of the fastest frameworks available.

Light: A framework with a long trail of dependencies becomes very difficult to incorporate in various environments due to the restrictions imposed by the dependencies. Falcon has only two dependencies: *six* (a Python 2 and 3 compatibility library, which facilitates the code bases to work on both Python 2 and 3 without requiring any changes) and *mimeparse* (which provides functionalities such as mime-type name parsing). This makes Falcon easier to test and deploy.